

ADVANCED CALCULUS AND PROBABILITY

Course Code: MS 8227
Shila, K

COLLEGE: Science and Technical Education
DEPARTMENT: Mathematics and Statistics

November 6, 2023



Mode of Assessment

Table 1: Mode of Assessment

CA & SE	WEEK	MARKS
Test 01	7 th	15%
Assignment	9 th	10%
Test 02	10 th	15%
Semester Exam	15 th	60%



Learning outcome



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- Ability to use a range of mathematical and statistical techniques and concepts:



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- Ability to use concepts of descriptive statistics and data presentation and interpretation.



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- Ability to apply knowledge of advanced calculus in solving physical and engineering problems
- Ability to apply knowledge Fourier in solving engineering problems



Course Contents



Course Contents

- Descriptive statistics



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- Descriptive statistics
- Basic probability



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- Descriptive statistics
- Basic probability
- Discrete Outcomes



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- Continuous outcomes



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- Fourier series and Transforms



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- Line and Multiple Integrals



Introduction



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Example of numerical data.



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- (ii) predict the future by analyzing the past events.
- (iii) find relationship between variables. For example, height and weight.



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- i The marks obtained by 10 students in a test. 6, 7, 5, 9, 1, 8, 8, 2, 7, 3 .

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example of raw data

- i The marks obtained by 10 students in a test. 6, 7, 5, 9, 1, 8, 8, 2, 7, 3 .
- ii Commercial transactions such as a purchase or return

Discrete data and continuous data

-



Discrete data and continuous data

- Discrete data assume only exact values



Discrete data and continuous data

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example discrete data



Figure 1: Number of students in different classes



Discrete data and continuous data

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Figure 1: Number of students in different classes

example of discrete data



Figure 2: Number of cars passing a certain point after every 15min.

Continuous data

- Data which assume intermediate values.



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example of continuous data



Figure 3: Weight s of students in a class



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Figure 3: Weight s of students in a class

example 2



Continuous data

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Figure 3: Weight s of students in a class

example 2

- Ages of pupils in a primary school etc.





Descriptive statistics



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A portion of a population selected for further analysis



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Statistic

A summary typically numerical of a variable over a sample or characteristics of a sample.



Range



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- The difference between greatest and smallest observations.



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Consider the numbers: 5, 6, 7, 8, 6, 5, 10, 7, 6



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- Range = 60%.

Frequency

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Example

Consider the numbers: 5, 6, 7, 8, 6, 5, 10, 7, 6

Number	5	6	7	8	10
Frequency	2	3	2	1	1



Frequency distribution table

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- A tabular arrangement which shows data and their corresponding frequencies.

- Relative frequency = $\frac{\text{Observed frequency}}{\text{Total frequency}}$

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- Relative frequency =
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Number	5	6	7	8	10
Frequency	2	3	2	1	1
Rel.Freq	$\frac{2}{9} = 22.22\%$	$\frac{3}{9} = 33.33\%$	$\frac{2}{9} = 22.22\%$	$\frac{1}{9} = 11.11\%$	$\frac{1}{9} = 11.11\%$

The relative frequency is generally expressed in percentage.



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Let's consider the interval $9 - 11$



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- The class interval 9 – 11 theoretically includes all marks (measurements) from 8.5 – 11.5.
- The number 8.5 and 11.5 are called boundaries or true class limits.
- 8.5 is the lower class boundary.
- 11.5 is the upper class boundary.
- How to get class boundary.
- Class boundary =
$$\frac{\text{U.c.l of nth class} + \text{L.c.l of (n + 1)th class}}{2}$$



Class size



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For the class interval 9 – 11 has class size of 3.



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- The mid-point of the class-interval.

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Marks	30-39	40-49	50-59	60-69	70-79
Frequency	10	14	26	20	18
Class boundary	29.5-39.5	39.5-49.5	49.5-59.5	59.5-69.5	69.5-79.5
Class mark	34.5	44.5	54.5	64.5	74.5

- Upper class boundary = $\frac{39 + 40}{2} = 39.5$
- Class Mark of 1st class interval = $\frac{30 + 39}{2} = 34.5$

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From the table above



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From the table above

$$\text{Class size} = 44.5 - 34.5 = 10.$$



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example

From the table above

Class size = $44.5 - 34.5 = 10$.

The class boundaries are mid-way between class-marks.

- Refer to table 3. u.c.b of class-mark 34.5 = $\frac{34.5 + 44.5}{2} = 39.5$

- l.c.b of 1st class-interval = $39.5 - 10 = 29.5$



Example



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If the class marks in the frequency distribution of lengths of certain leaves are 128, 137, 146, 155, 164, 173 and 182. Find



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If the class marks in the frequency distribution of lengths of certain leaves are 128, 137, 146, 155, 164, 173 and 182. Find

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- b) the class boundaries.



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If the class marks in the frequency distribution of lengths of certain leaves are 128, 137, 146, 155, 164, 173 and 182. Find

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- c) the class limits assuming that the length were measured to the nearest millimetre.

Solution



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If the class marks in the frequency distribution of lengths of certain leaves are 128, 137, 146, 155, 164, 173 and 182. Find

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Solution

- a) Class size = $137 - 128 = 9$.



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- a) Class size = $137 - 128 = 9$.
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a) Class size = $137 - 128 = 9$.

b) Class-boundaries

$$\text{u.c.b of 1}^{st} \text{ class mark} = \frac{128 + 137}{2} = 132.5$$



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Solution

a) Class size = $137 - 128 = 9$.

b) Class-boundaries

$$\text{u.c.b of } 1^{\text{st}} \text{ class mark} = \frac{128 + 137}{2} = 132.5$$

$$\text{l.c.b of } 1^{\text{st}} \text{ class mark} = 132.5 - 9 = 123.5$$



Solution



Solution

(c) Solution



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Class Marks	Class boundary	Class limits	Frequency
128	123.5 - 132.5	124 – 132	1
137	132.5 – 141.5	133 – 141	1
146	141.5 – 150.5	142 – 150	1
155	150.5 - 159.5	151 – 159	1
164	159.5 -168.5	160 – 168	1
173	168.5 - 177.5	169 – 177	1
182	177.5 – 186.5	178 – 186	1



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164	159.5 -168.5	160 – 168	1
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- Data summarized in classes, for example 124 – 132, is called grouped data.



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When discrete data is arranged into classes certain original information is lost.



Continuous frequency distribution



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In such a case we form class-intervals as follows:

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Age in years

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15 or more but < 20 .

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Age in years

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15 or more but < 20 .

20 or more but < 25 .

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In such a case we form class-intervals as follows:

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15 or more but < 20 .

20 or more but < 25 .

25 or more but < 30 .

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Example

In such a case we form class-intervals as follows:

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15 or more but < 20 .

20 or more but < 25 .

25 or more but < 30 .

And so on.



This way all persons with any fraction of age are included in one group or the other.

Example



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Example

For practical purpose we write the above classes as

15 - 20

20 - 25

25 - 30

30 - 35

and so on.

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For practical purpose we write the above classes as

15 - 20

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The following are the results in mathematics test of twenty students.



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For practical purpose we write the above classes as

15 - 20

20 - 25

25 - 30

30 - 35

and so on.

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The following are the results in mathematics test of twenty students.

27, 21, 24, 27, 31, 40, 45, 46, 50, 40, 38, 29, 49, 48, 35, 34, 25, 26, 35, 25.

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Example

For practical purpose we write the above classes as

15 - 20

20 - 25

25 - 30

30 - 35

and so on.

Example

The following are the results in mathematics test of twenty students.

27, 21, 24, 27, 31, 40, 45, 46, 50, 40, 38, 29, 49, 48, 35, 34, 25, 26, 35, 25.

Prepare the frequency distribution by using class-Intervals, 21 - 25, 26 - 30, 31 - 35,.....



example



example

The following are results of English examination of 50 students in ABC school.



example

The following are results of English examination of 50 students in ABC school.

21, 23, 48, 54, 64, 77, 68, 59, 31, 46, 33, 53, 61, 71, 82, 75, 61, 64, 34, 25, 26, 31, 32, 36, 48, 45, 45, 44, 55, 54, 60, 67, 71, 74, 37, 38, 34, 29, 40, 41, 26, 42, 78, 80, 86, 90, 43, 27, 45, 67.



example

The following are results of English examination of 50 students in ABC school.

21, 23, 48, 54, 64, 77, 68, 59, 31, 46, 33, 53, 61, 71, 82, 75, 61, 64, 34, 25, 26, 31, 32, 36, 48, 45, 45, 44, 55, 54, 60, 67, 71, 74, 37, 38, 34, 29, 40, 41, 26, 42, 78, 80, 86, 90, 43, 27, 45, 67.

Draw a frequency distribution by using class marks 23, 28, 33,

Solution

- Given the class marks 23, 28, 33,
- Upper class boundary of class mark 23 = $\frac{23 + 28}{2} = 25.5$
- Class-size = $28 - 23 = 5$
- Lower class boundary of class mark 23 = $25.5 - 5 = 20.5$



solution



solution

Class Interval	Frequency	Class mark
21-25	3	23
26-30	4	28
31-35	6	33
36-40	4	38
42-45	7	43
46-50	3	48
51-55	4	53
56-60	2	58
61-65	4	63
66-70	3	68
71-75	4	73
76-80	3	78
81-85	1	83
86-90	2	88



Example



Example

The data below gives the time in minutes it takes a commuter to drive to work for a period lasting 50 days.



Example

The data below gives the time in minutes it takes a commuter to drive to work for a period lasting 50 days.

25, 27, 43, 23, 39, 33, 29, 26, 34, 22, 28, 30, 39, 32, 25, 27, 28, 28, 27, 35, 28, 46, 24, 24, 22, 31, 28, 27, 31, 22, 32, 36, 22, 26, 34, 30, 37, 35, 42, 25, 37, 30, 27, 31, 30, 48, 28, 24, 25.



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The data below gives the time in minutes it takes a commuter to drive to work for a period lasting 50 days.

25, 27, 43, 23, 39, 33, 29, 26, 34, 22, 28, 30, 39, 32, 25, 27, 28, 28, 27, 35, 28, 46, 24, 24, 22, 31, 28, 27, 31, 22, 32, 36, 22, 26, 34, 30, 37, 35, 42, 25, 37, 30, 27, 31, 30, 48, 28, 24, 25.

Construct the freq. distr.table having six classes for which 20 is the lower limit of a first class and 49 is the upper limit of the sixth class.

Example



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The data below gives the time in minutes it takes a commuter to drive to work for a period lasting 50 days.

25, 27, 43, 23, 39, 33, 29, 26, 34, 22, 28, 30, 39, 32, 25, 27, 28, 28, 27, 35, 28, 46, 24, 24, 22, 31, 28, 27, 31, 22, 32, 36, 22, 26, 34, 30, 37, 35, 42, 25, 37, 30, 27, 31, 30, 48, 28, 24, 25.

Construct the freq. distr.table having six classes for which 20 is the lower limit of a first class and 49 is the upper limit of the sixth class.

Example

Make a frequency table using five classes for the following data.



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The data below gives the time in minutes it takes a commuter to drive to work for a period lasting 50 days.

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Construct the freq. distr.table having six classes for which 20 is the lower limit of a first class and 49 is the upper limit of the sixth class.

Example

Make a frequency table using five classes for the following data.

60, 25, 29, 25, 40, 35, 50, 80, 80, 90, 80, 55, 70, 40, 100, 52, 30, 70, 45, 60.



Assignment

- Discuss the methods of data collection.
- Write short notes on histogram, frequency polygon, cumulative frequency curve(ogive), pie and bar charts. Show how to apply the graphical data representations by solving one question each. Date of submission.
- Describe sampling techniques. Date of submission 13/11/2023.



Measures of central tendency

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- A measure of central tendency indicates the location or the general position of the distribution on the x-axis.



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- Each in a different sense tells what value of the data is clustered around
- A measure of central tendency indicates the location or the general position of the distribution on the x-axis.
- It is also known as a measure of location or position.



Measures of central tendency cont...

These includes;



Measures of central tendency cont...

These includes;

- Mean



Measures of central tendency cont...

These includes;

- Mean
- Mode



Measures of central tendency cont...

These includes;

- Mean
- Mode
- Median





Arithmetic Mean



Arithmetic Mean

- Arithmetic mean $\bar{x}$ is the average of the given data (values)



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Un grouped data



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- Arithmetic mean $\bar{x}$ of a set of $x_1, x_2, x_3, \dots, x_n$ observations is their sum divided by the number of observations. i.e.



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$$\bar{x} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{N}$$

Where $N = 1 + 1 + 1 + \dots + 1 = n$



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$$\bar{x} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{N}$$

Where $N = 1 + 1 + 1 + \dots + 1 = n$

- If f_i is the frequency of the observations $x_i = 1, 2, \dots, n$ then

$$\bar{x} = \frac{f_1 x_1 + f_2 x_2 + \dots + f_n x_n}{f_1 + f_2 + \dots + f_n}$$



Mean Cont...

$$\blacksquare \bar{x} = \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i} = \frac{\sum fx}{\sum f} = \frac{\sum fx}{N}$$

Example



Mean Cont...

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Example

Find mean of the set of numbers 63, 65, 67, 68, 69, 70, 71, 72, 73, 74, 75.



Mean Cont...

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Example

Find mean of the set of numbers 63, 65, 67, 68, 69, 70, 71, 72, 73, 74, 75.

Solution

$$\bar{x} = \frac{63 + 65 + 67 + 68 + 69 + 70 + 71 + 72 + 73 + 74 + 75}{11}$$



Mean Cont...

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$$\bar{x} = \frac{63 + 65 + 67 + 68 + 69 + 70 + 71 + 72 + 73 + 74 + 75}{11}$$

$$= 69.73$$



Mean cont...



Mean cont...

Example

Mean cont...

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Mean cont...

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Mean cont...

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Marks	0-10	10-20	20-30	30-40	40-50	50-60
Frequency	12	18	27	20	17	6

Solution

Mean cont...

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Marks	Class mark (x)	Frequency (f)	fx
0-10	5	12	60
10-20	15	18	270
20-30	25	27	675
30-40	35	20	700
40-50	45	17	495
50-60	55	6	330
		$N = \sum f = 100$	$\sum fx = 2800$

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$$\bar{x} = \frac{\sum fx}{N} = \frac{2800}{100}$$

Mean cont...

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		$N = \sum f = 100$	$\sum fx = 2800$

$$\begin{aligned}\bar{x} &= \frac{\sum fx}{N} = \frac{2800}{100} \\ &= 28\end{aligned}$$

Mean cont...

ASSUMED MEAN METHOD



Mean cont...

ASSUMED MEAN METHOD

- Suppose we have a set of observations $x_1, x_2, \dots, x_n$ with their frequencies $f_1, f_2, \dots, f_n$
- Let $\bar{x}_a$ = Assumed mean
- Let $d_i = x_i - \bar{x}_a$ (deviation of observation x_i from $\bar{x}_a$) where $i = 1, 2, \dots, n$

- Now,
$$\sum_{i=1}^n f_i d_i = \sum_{i=1}^n (x_i - \bar{x}_a)$$

$$= \sum f_i x_i - \bar{x}_a \sum f_i$$

- Suppose we divide both sides by $(N = \sum f)$ we get

- $$\frac{\sum fd}{N} = \frac{\sum fx}{N} = \frac{\bar{x}_a \sum f}{N}$$

- $\bar{d} = \bar{x} - \bar{x}_a$, Thus $\bar{x} = \bar{d} + \bar{x}_a$,



Mean cont...

Example



Mean cont...

Example

- Choose the central value to be the assumed mean.



Mean cont...

Example

- Choose the central value to be the assumed mean.
- Find the mean of the set of numbers



Mean cont...

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- Find the mean of the set of numbers
63, 65, 67, 68, 69, 70, 71, 72, 73, 74, 75.



Mean cont...

Example

- Choose the central value to be the assumed mean.
- Find the mean of the set of numbers 63, 65, 67, 68, 69, 70, 71, 72, 73, 74, 75. by assumed mean method.



Mean cont...

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by assumed mean method.
- Solution



Mean cont...

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- Choose the central value to be the assumed mean.

- Find the mean of the set of numbers
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by assumed mean method.

- Solution

Let take $\bar{x}_a = 70$ as the assumed mean, then we have



Mean cont...

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Numbers(x)	63	65	67	68	69	70	71	72	73	74	75
$d_i = x_i - 70$	-7	-5	-3	-2	-1	0	1	2	3	4	5



Mean cont...

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Mean cont...

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Therefore $\bar{x} = 69.73$



Example



Example

- Given the table, find arithmetic mean by



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- Given the table, find arithmetic mean by
(a) direct method



Example

- Given the table, find arithmetic mean by
 - (a) direct method
 - (b) assumed mean method



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Class interval	Frequency
0-8	8
8 -16	7
16-24	16
24-32	24
32-40	15
40-48	7

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- The following data set represents the age distribution of a sample of 72 women having multiple delivery births in 2018.

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- The following data set represents the age distribution of a sample of 72 women having multiple delivery births in 2018.

Age(Years)	15-19	20-24	25-29	30-34	35-39	40-44
Numbers	2	5	17	28	17	3

Example

- Given the table, find arithmetic mean by
 - direct method
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Age(Years)	15-19	20-24	25-29	30-34	35-39	40-44
Numbers	2	5	17	28	17	3

- Find the mean age of the women in years.
 - By direct method
 - By using Assumed mean method

Mean cont...

COMBINED MEAN



Mean cont...

COMBINED MEAN

- Let n_1 = Number of observation of group one



Mean cont...

COMBINED MEAN

- Let n_1 = Number of observation of group one
- n_2 = Number of observation of group two



Mean cont...

COMBINED MEAN

- Let n_1 = Number of observation of group one
- n_2 = Number of observation of group two
- $\bar{x}_1$ = Arithmetic mean of group one



Mean cont...

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- Let n_1 = Number of observation of group one
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Mean cont...

COMBINED MEAN

- Let n_1 = Number of observation of group one
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Mean cont...

COMBINED MEAN

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- n_2 = Number of observation of group two
- $\bar{x}_1$ = Arithmetic mean of group one
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Then, the combined mean is given by

- $$\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$



Mean cont...

proof

- Let $x_{11}, x_{12}, x_{13}, \dots, x_{1n}$ = be the observation of the group one
- Let $x_{21}, x_{22}, x_{23}, \dots, x_{2n}$ = be the observation of the group two



Mean cont...

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- Let $x_{11}, x_{12}, x_{13}, \dots, x_{1n}$ be the observation of the group one
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- $\bar{x}_1 = \frac{f_{11}x_{11} + f_{12}x_{12} + f_{13}x_{13} + \dots + f_{1n}x_{1n}}{f_{11} + f_{12} + f_{13} + \dots + f_{1n}}$ or

Mean cont...

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Mean cont...

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- $f_{21}x_{21} + f_{22}x_{22} + f_{23}x_{23} + \dots + f_{2n}x_{2n} = n_2 \bar{x}_2 \dots \dots \dots (2)$

Mean cont...

proof

- Let $x_{11}, x_{12}, x_{13}, \dots, x_{1n}$ = be the observation of the group one
- Let $x_{21}, x_{22}, x_{23}, \dots, x_{2n}$ = be the observation of the group two
- $\bar{x}_1 = \frac{f_{11}x_{11} + f_{12}x_{12} + f_{13}x_{13} + \dots + f_{1n}x_{1n}}{f_{11} + f_{12} + f_{13} + \dots + f_{1n}}$ or
- $f_{11}x_{11} + f_{12}x_{12} + f_{13}x_{13} + \dots + f_{1n}x_{1n} = n_1 \bar{x}_1 \dots \dots \dots (1)$
- $\bar{x}_2 = \frac{f_{21}x_{21} + f_{22}x_{22} + f_{23}x_{23} + \dots + f_{2n}x_{2n}}{f_{21} + f_{22} + f_{23} + \dots + f_{2n}}$ or
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- combined mean

Mean cont...

proof

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$$f_{21}x_{21} + f_{22}x_{22} + f_{23}x_{23} + \dots + f_{2n}x_{2n} = n_2 \bar{x}_2 \dots \dots \dots (2)$$

- combined mean

$$\bar{x} = \frac{f_{11}x_{11} + f_{21}x_{21} + f_{12}x_{12} + f_{22}x_{22} + \dots + f_{1n}x_{1n} + f_{2n}x_{2n}}{f_{11} + f_{21} + f_{12} + f_{22} + \dots + f_{1n} + f_{2n}} \dots (3)$$

Mean cont...



Mean cont...

- Using (1) and (2) in (3) Results to;



Mean cont...

- Using (1) and (2) in (3) Results to;

- $$\bar{X} = \frac{n_1\bar{X}_1 + n_2\bar{X}_2}{n_1 + n_2}$$



Mean cont...

- Using (1) and (2) in (3) Results to;

- $\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$

- in general $\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2 + \cdots + n_n\bar{x}_n}{n_1 + n_2 + \cdots + n_n}$

Example



Mean cont...

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- $$\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$$

- in general
$$\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2 + \cdots + n_n\bar{x}_n}{n_1 + n_2 + \cdots + n_n}$$

Example

- The mean salary for a group of 40 female workers is \$ 5200 per month and for a group of 60 male workers is \$ 6800 per month. What is the combined mean salary.



Mean cont...

Solution



Mean cont...

Solution

- Let n_1 = Number of female workers



Mean cont...

Solution

- Let n_1 = Number of female workers
- n_2 = Number of male workers



Mean cont...

Solution

- Let n_1 = Number of female workers
- n_2 = Number of male workers
- $\bar{x}_1$ = Mean salary for female workers



Mean cont...

Solution

- Let n_1 = Number of female workers
- n_2 = Number of male workers
- $\bar{x}_1$ = Mean salary for female workers
- $\bar{x}_2$ = Mean salary for male workers



Mean cont...

Solution

- Let n_1 = Number of female workers
- n_2 = Number of male workers
- $\bar{x}_1$ = Mean salary for female workers
- $\bar{x}_2$ = Mean salary for male workers

Then, the combined mean is given by



Mean cont...

Solution

- Let n_1 = Number of female workers
- n_2 = Number of male workers
- $\bar{x}_1$ = Mean salary for female workers
- $\bar{x}_2$ = Mean salary for male workers

Then, the combined mean is given by



Mean cont...

Solution

- Let n_1 = Number of female workers
- n_2 = Number of male workers
- $\bar{x}_1$ = Mean salary for female workers
- $\bar{x}_2$ = Mean salary for male workers

Then, the combined mean is given by

$$\begin{aligned}
 \bar{x} &= \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2} \\
 &= \frac{40(5200) + 60(6800)}{40 + 60} \\
 &= 6160
 \end{aligned}$$



Mean cont...

Solution

- Let n_1 = Number of female workers
- n_2 = Number of male workers
- $\bar{x}_1$ = Mean salary for female workers
- $\bar{x}_2$ = Mean salary for male workers

Then, the combined mean is given by

$$\begin{aligned}
 \bar{x} &= \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2} \\
 &= \frac{40(5200) + 60(6800)}{40 + 60} \\
 &= 6160
 \end{aligned}$$

- The combined mean salary is \$ 6160



Mean cont...

Example

- The arithmetic mean of 100 items were recorded as 40, but later on it was discovered that one observation 40 was wrong copied down as 50. Find the correct mean.



Mean cont...

Example

- The arithmetic mean of 100 items were recorded as 40, but later on it was discovered that one observation 40 was wrong copied down as 50. Find the correct mean.

Solution



Mean cont...

Example

- The arithmetic mean of 100 items were recorded as 40, but later on it was discovered that one observation 40 was wrong copied down as 50. Find the correct mean.

Solution

Given:



Mean cont...

Example

- The arithmetic mean of 100 items were recorded as 40, but later on it was discovered that one observation 40 was wrong copied down as 50. Find the correct mean.

Solution

Given:

Number of items = 100.



Mean cont...

Example

- The arithmetic mean of 100 items were recorded as 40, but later on it was discovered that one observation 40 was wrong copied down as 50. Find the correct mean.

Solution

Given:

Number of items = 100.

Incorrect mean = 40.



Mean cont...

Example

- The arithmetic mean of 100 items were recorded as 40, but later on it was discovered that one observation 40 was wrong copied down as 50. Find the correct mean.

Solution

Given:

Number of items = 100.

Incorrect mean = 40.

Incorrect value = 50.



Mean cont...

Example

- The arithmetic mean of 100 items were recorded as 40, but later on it was discovered that one observation 40 was wrong copied down as 50. Find the correct mean.

Solution

Given:

Number of items = 100.

Incorrect mean = 40.

Incorrect value = 50.

Correct value = 40.



Mean cont...

Example

- The arithmetic mean of 100 items were recorded as 40, but later on it was discovered that one observation 40 was wrong copied down as 50. Find the correct mean.

Solution

Given:

Number of items = 100.

Incorrect mean = 40.

Incorrect value = 50.

Correct value = 40.

Required the correct mean.

- correct mean = $\frac{\text{correct } \sum x}{n}$

- but, $\text{correct } \sum x = \text{Incorrect } \sum x - \text{incorrect value} + \text{correct value}$

- $= n(\text{incorrect } \bar{x}) - \text{incorrect value} + \text{correct value}.$



Mean cont...

- 3990
- Now, the correct mean = $\frac{3990}{100} = 39.9$
- Therefore, the correct mean = 39.9

Example



Mode



Mode

- Mode is a value which occurs most frequently in the given distribution.

Mode for ungrouped data



Mode

- Mode is a value which occurs most frequently in the given distribution.

Mode for ungrouped data

- Mode for ungrouped data is obtained by checking or inspecting data which occurs mostly.

Mode for grouped data

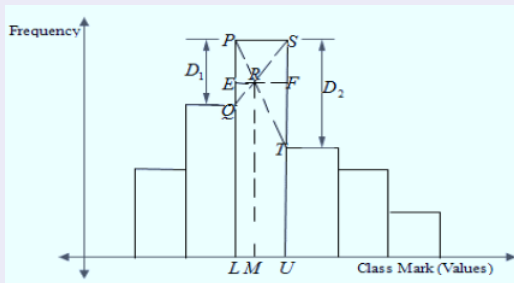


Figure 4: Consider the histogram below

Mode cont...



Mode cont...

- From the histogram above



Mode cont...

- From the histogram above
- ΔPRQ is similar to ΔSRT



Mode cont...

- From the histogram above
- $\triangle PRQ$ is similar to $\triangle SRT$
- $\frac{ER}{FR} = \frac{PQ}{ST}$



Mode cont...

- From the histogram above

- $\triangle PRQ$ is similar to $\triangle SRT$

- $$\frac{ER}{FR} = \frac{PQ}{ST}$$

- $$\frac{M-L}{U-M} = \frac{D_1}{D_2}$$



Mode cont...

- From the histogram above
- $\triangle PRQ$ is similar to $\triangle SRT$
- $\frac{ER}{FR} = \frac{PQ}{ST}$
- $\frac{M-L}{U-M} = \frac{D_1}{D_2}$
- $(M-L)D_2 = (U-M)D_1$



Mode cont...

- From the histogram above
- $\triangle PRQ$ is similar to $\triangle SRT$
- $\frac{ER}{FR} = \frac{PQ}{ST}$
- $\frac{M-L}{U-M} = \frac{D_1}{D_2}$
- $(M-L)D_2 = (U-M)D_1$
- $MD_2 - LD_2 = UD_1 - MD_1$



Mode cont...

- From the histogram above

- $\triangle PRQ$ is similar to $\triangle SRT$

- $$\frac{ER}{FR} = \frac{PQ}{ST}$$

- $$\frac{M-L}{U-M} = \frac{D_1}{D_2}$$

- $$(M-L)D_2 = (U-M)D_1$$

- $$MD_2 - LD_2 = UD_1 - MD_1$$

- $$MD_2 + MD_1 = UD_1 + LD_2$$



Mode cont...

- From the histogram above
- $\triangle PRQ$ is similar to $\triangle SRT$
- $\frac{ER}{FR} = \frac{PQ}{ST}$
- $\frac{M-L}{U-M} = \frac{D_1}{D_2}$
- $(M-L)D_2 = (U-M)D_1$
- $MD_2 - LD_2 = UD_1 - MD_1$
- $MD_2 + MD_1 = UD_1 + LD_2$
- $M(D_1 + D_2) = UD_1 + LD_2 \dots (i)$



Mode cont...

- From the histogram above
- $\triangle PRQ$ is similar to $\triangle SRT$
- $\frac{ER}{FR} = \frac{PQ}{ST}$
- $\frac{M-L}{U-M} = \frac{D_1}{D_2}$
- $(M-L)D_2 = (U-M)D_1$
- $MD_2 - LD_2 = UD_1 - MD_1$
- $MD_2 + MD_1 = UD_1 + LD_2$
- $M(D_1 + D_2) = UD_1 + LD_2 \dots (i)$
- But, class size $C = U - L, U = C + L \dots (ii)$



Mode cont...

- From the histogram above
- $\triangle PRQ$ is similar to $\triangle SRT$
- $\frac{ER}{FR} = \frac{PQ}{ST}$
- $\frac{M-L}{U-M} = \frac{D_1}{D_2}$
- $(M-L)D_2 = (U-M)D_1$
- $MD_2 - LD_2 = UD_1 - MD_1$
- $MD_2 + MD_1 = UD_1 + LD_2$
- $M(D_1 + D_2) = UD_1 + LD_2 \dots (i)$
- But, class size $C = U - L, U = C + L \dots (ii)$
- substitute equation (ii) into (i)



Mode cont...



Mode cont...

$$\blacksquare M(D_1 + D_2) = (C + L)D_1 + LD_2$$



Mode cont...

- $M(D_1 + D_2) = (C + L)D_1 + LD_2$
- $M(D_1 + D_2) = CD_1 + LD_1 + LD_2$



Mode cont...

- $M(D_1 + D_2) = (C + L)D_1 + LD_2$
- $M(D_1 + D_2) = CD_1 + LD_1 + LD_2$
- $M(D_1 + D_2) = +L(D_1 + D_2) + CD_1 \dots \text{(iii)}$



Mode cont...

- $M(D_1 + D_2) = (C + L)D_1 + LD_2$
- $M(D_1 + D_2) = CD_1 + LD_1 + LD_2$
- $M(D_1 + D_2) = +L(D_1 + D_2) + CD_1 \dots$ (iii)
- divide by $(D_1 + D_2)$ through out equation (iii)



Mode cont...

- $M(D_1 + D_2) = (C + L)D_1 + LD_2$
- $M(D_1 + D_2) = CD_1 + LD_1 + LD_2$
- $M(D_1 + D_2) = +L(D_1 + D_2) + CD_1 \dots (iii)$
- divide by $(D_1 + D_2)$ through out equation (iii)
- $\text{Mode}(M) = L + \left(\frac{D_1}{D_1 + D_2} \right) C$ where;
- L = Lower class boundary
- D_1 = The difference between the frequency of modal class and the frequency of the class before modal class
- D_2 = The difference between the frequency of modal class and the frequency of the class after modal class
- C = Class size



Example

- Find the mode of the following scores 40,30,25,10,40,19,40

Solution

- The mode is 40

Example

- Calculate the mean and mode of the following distribution below



Example

- Find the mode of the following scores 40,30,25,10,40,19,40

Solution

- The mode is 40

Example

- Calculate the mean and mode of the following distribution below



Example

- Find the mode of the following scores 40,30,25,10,40,19,40

Solution

- The mode is 40

Example

- Calculate the mean and mode of the following distribution below



Example

- Find the mode of the following scores 40,30,25,10,40,19,40

Solution

- The mode is 40

Example

- Calculate the mean and mode of the following distribution below

Class interval	Frequency
0-30	4
30 -60	7
60-90	14
90-120	11
120-150	2



Solution

- Mean



Solution

- Mean



Solution

■ Mean



Solution

■ Mean

Class interval	Frequency (f)	x	fx
0-30	4	15	60
30 -60	7	45	315
60-90	14	75	1050
90-120	11	105	1155
120-150	2	135	270

Solution

■ Mean

Class interval	Frequency (f)	x	fx
0-30	4	15	60
30 -60	7	45	315
60-90	14	75	1050
90-120	11	105	1155
120-150	2	135	270

■ $\sum f = 38$

Solution

■ Mean

Class interval	Frequency (f)	x	fx
0-30	4	15	60
30 -60	7	45	315
60-90	14	75	1050
90-120	11	105	1155
120-150	2	135	270

■ $\sum f = 38$

■ $\sum fx = 2850$



Solution

■ Mean

Class interval	Frequency (f)	x	fx
0-30	4	15	60
30 -60	7	45	315
60-90	14	75	1050
90-120	11	105	1155
120-150	2	135	270

■ $\sum f = 38$

■ $\sum fx = 2850$

■ Mean ($\bar{x}$) = $\frac{\sum fx}{\sum f} = \frac{2850}{38} = 75$

Solution

■ Mean

Class interval	Frequency (f)	x	fx
0-30	4	15	60
30 -60	7	45	315
60-90	14	75	1050
90-120	11	105	1155
120-150	2	135	270

■ $\sum f = 38$

■ $\sum fx = 2850$

■ Mean ($\bar{x}$) = $\frac{\sum fx}{\sum f} = \frac{2850}{38} = 75$

■ Modal class 60-90, $L=60$, $C=30$, $D_1=14-7=7$ and $D_2=14-11=3$



Solution

■ Mean

Class interval	Frequency (f)	x	fx
0-30	4	15	60
30 -60	7	45	315
60-90	14	75	1050
90-120	11	105	1155
120-150	2	135	270

■ $\sum f = 38$

■ $\sum fx = 2850$

■ Mean ($\bar{x}$) = $\frac{\sum fx}{\sum f} = \frac{2850}{38} = 75$

■ Modal class 60-90, $L=60$, $C=30$, $D_1=14-7=7$ and $D_2=14-11=3$

■ Mode (M) = $L + \left(\frac{D_1}{D_1 + D_2} \right) C = 60 + \left(\frac{7}{7+3} \right) 30 = 81$



Median

Median



Median

Median

- Median is the middle value of the given distribution when arranged in ascending or descending order.

Median from ungrouped data



Median

Median

- Median is the middle value of the given distribution when arranged in ascending or descending order.

Median from ungrouped data

- Case I: When the number of observations (n) is odd, then median value is $(\frac{n+1}{2})^{th}$ observation.



Median

Median

- Median is the middle value of the given distribution when arranged in ascending or descending order.

Median from ungrouped data

- Case I: When the number of observations (n) is odd, then median value is $(\frac{n+1}{2})^{th}$ observation.
- Case II: When the number of observations (n) is even, the median is the average of the two numbers $(\frac{n}{2})^{th}$ and $(\frac{n}{2} + 1)^{th}$ observation.



Median cont...

Median cont...



Median cont...

Median cont...

- Median for grouped data is given by the formula



Median cont...

Median cont...

- Median for grouped data is given by the formula

- Median = $L + \left(\frac{\frac{N}{2} - \sum f_b}{f_w} \right) c$

Where,



Median cont...

Median cont...

- Median for grouped data is given by the formula

- Median = $L + \left(\frac{\frac{N}{2} - \sum f_b}{f_w} \right) c$

Where,

L = Lower class boundary of median class



Median cont...

Median cont...

- Median for grouped data is given by the formula

- Median = $L + \left(\frac{\frac{N}{2} - \sum f_b}{f_w} \right) c$

Where,

L= Lower class boundary a of median class

N=Sum of the frequencies



Median cont...

Median cont...

- Median for grouped data is given by the formula

- Median = $L + \left(\frac{\frac{N}{2} - \sum f_b}{f_w} \right) c$

Where,

L= Lower class boundary a of median class

N=Sum of the frequencies

$\sum f_b$ = Sum of frequency before the frequency of median class



Median cont...

Median cont...

- Median for grouped data is given by the formula

- Median = $L + \left(\frac{\frac{N}{2} - \sum f_b}{f_w} \right) c$

Where,

L = Lower class boundary of median class

N = Sum of the frequencies

$\sum f_b$ = Sum of frequency before the frequency of median class

f_w = Frequency within the median class



Median cont...

Median cont...

- Median for grouped data is given by the formula

- Median = $L + \left(\frac{\frac{N}{2} - \sum f_b}{f_w} \right) c$

Where,

L= Lower class boundary a of median class

N=Sum of the frequencies

$\sum f_b$ = Sum of frequency before the frequency of median class

f_w = Frequency within the median class

c=Class size

Example

Median cont...

Median cont...

- Median for grouped data is given by the formula

- Median = $L + \left(\frac{\frac{N}{2} - \sum f_b}{f_w} \right) c$

Where,

L= Lower class boundary a of median class

N=Sum of the frequencies

$\sum f_b$ = Sum of frequency before the frequency of median class

f_w = Frequency within the median class

c=Class size

Example

- Find the median of the following scores
19,27,19,18,28,35

Median cont...

Example



Median cont...

Example

- Find the median of the value.



Median cont...

Example

- Find the median of the value.
(i) 25, 20, 15, 14, 18.



Median cont...

Example

- Find the median of the value.
 - (i) 25, 20, 15, 14, 18.
 - (ii) 16, 17, 20, 12, 30, 38.



Median cont...

Example

- Find the median of the value.
 - (i) 25, 20, 15, 14, 18.
 - (ii) 16, 17, 20, 12, 30, 38.
- Given the following distribution table,



Median cont...

Example

- Find the median of the value.
 - (i) 25, 20, 15, 14, 18.
 - (ii) 16, 17, 20, 12, 30, 38.
- Given the following distribution table,



Median cont...

Example

- Find the median of the value.
 - (i) 25, 20, 15, 14, 18.
 - (ii) 16, 17, 20, 12, 30, 38.
- Given the following distribution table,

--	--



Median cont...

Example

- Find the median of the value.
 - (i) 25, 20, 15, 14, 18.
 - (ii) 16, 17, 20, 12, 30, 38.
- Given the following distribution table,

Class interval	Frequency
21-25	5
26-30	4
31-35	12
36-40	6
41-45	3
46-50	8



Median cont...

Example

- Find the median of the value.
 - (i) 25, 20, 15, 14, 18.
 - (ii) 16, 17, 20, 12, 30, 38.
- Given the following distribution table,

Class interval	Frequency
21-25	5
26-30	4
31-35	12
36-40	6
41-45	3
46-50	8

Calculate



Median cont...

Example

- Find the median of the value.
 - (i) 25, 20, 15, 14, 18.
 - (ii) 16, 17, 20, 12, 30, 38.
- Given the following distribution table,

Class interval	Frequency
21-25	5
26-30	4
31-35	12
36-40	6
41-45	3
46-50	8

Calculate

(a) Mean by assumed method



Median cont...

Example

- Find the median of the value.
 - (i) 25, 20, 15, 14, 18.
 - (ii) 16, 17, 20, 12, 30, 38.
- Given the following distribution table,

Class interval	Frequency
21-25	5
26-30	4
31-35	12
36-40	6
41-45	3
46-50	8

Calculate

- (a) Mean by assumed method
- (b) Mode (c) Median



Solution



Solution



Solution



Solution

Class interval	f	c.f	x	$d = x - \bar{x}_a$	fd
21-25	5	5	23	-10	-50
26-30	4	9	28	-5	-20
31-35	12	21	33	0	0
36-40	6	27	38	5	30
41-45	3	30	43	10	30
46-50	8	38	48	15	120
	$\Sigma f = 38$				110

■ $\bar{x}_a = 33$

■ $c=5$

■ $\Sigma fd = 110$

■ $\Sigma f = 38$

■ $\bar{d} = \frac{\Sigma fd}{\Sigma f} = \frac{110}{38}$

(a) Therefore, Mean $\bar{x} = \bar{x}_a + \bar{d} = 33 + 2.89 = 35.89$





$$(b) \text{ Mode } (M) = L + \left(\frac{D_1}{D_1 + D_2} \right) C$$



$$(b) \text{ Mode } (M) = L + \left(\frac{D_1}{D_1 + D_2} \right) C$$

- The modal class is 31-35



$$(b) \text{ Mode } (M) = L + \left(\frac{D_1}{D_1 + D_2} \right) C$$

- The modal class is 31-35
- $L = 31 - 0.5 = 30.5$



$$(b) \text{ Mode } (M) = L + \left(\frac{D_1}{D_1 + D_2} \right) C$$

- The modal class is 31-35
- $L = 31 - 0.5 = 30.5$
- $D_1 = 12 - 4 = 8$



$$(b) \text{ Mode } (M) = (M) = L + \left(\frac{D_1}{D_1 + D_2} \right) C$$

- The modal class is 31-35
- $L = 31 - 0.5 = 30.5$
- $D_1 = 12 - 4 = 8$
- $D_2 = 12 - 6 = 6$



$$(b) \text{ Mode } (M) = (M) = L + \left(\frac{D_1}{D_1 + D_2} \right) C$$

- The modal class is 31-35
- $L = 31 - 0.5 = 30.5$
- $D_1 = 12 - 4 = 8$
- $D_2 = 12 - 6 = 6$
- $c = 5$



$$(b) \text{ Mode } (M) = L + \left(\frac{D_1}{D_1 + D_2} \right) C$$

■ The modal class is 31-35

■ $L = 31 - 0.5 = 30.5$

■ $D_1 = 12 - 4 = 8$

■ $D_2 = 12 - 6 = 6$

■ $c = 5$

■ $\text{Mode } (M) = L + \left(\frac{D_1}{D_1 + D_2} \right) C = 30.5 + \left(\frac{8}{8 + 6} \right) 5 = 33.36$



(c) Median

(c)Median

$$\blacksquare \text{ Median} = L + \left(\frac{\frac{N}{2} - \sum f_b}{f_w} \right) c$$

Where,

(c)Median

$$\blacksquare \text{ Median} = L + \left(\frac{\frac{N}{2} - \sum f_b}{f_w} \right) c$$

Where,

(c)Median

- Median = $L + \left(\frac{\frac{N}{2} - \sum f_b}{f_w} \right) c$

Where,

- L = 30.5, median class 31-35

(c)Median

- Median = $L + \left(\frac{\frac{N}{2} - \sum f_b}{f_w} \right) c$

Where,

- L = 30.5, median class 31-35

(c)Median

$$\blacksquare \text{ Median} = L + \left(\frac{\frac{N}{2} - \sum f_b}{f_w} \right) c$$

Where,

- $L = 30.5$, median class 31-35
- $N = 38$

(c)Median

$$\blacksquare \text{ Median} = L + \left(\frac{\frac{N}{2} - \sum f_b}{f_w} \right) c$$

Where,

- $L = 30.5$, median class 31-35
- $N = 38$

(c)Median

$$\blacksquare \text{ Median} = L + \left(\frac{\frac{N}{2} - \sum f_b}{f_w} \right) c$$

Where,

- $L = 30.5$, median class 31-35
- $N = 38$
- $\sum f_b = 9$

(c)Median

$$\blacksquare \text{ Median} = L + \left(\frac{\frac{N}{2} - \sum f_b}{f_w} \right) c$$

Where,

- $L = 30.5$, median class 31-35
- $N = 38$
- $\sum f_b = 9$

(c)Median

$$\blacksquare \text{ Median} = L + \left(\frac{\frac{N}{2} - \sum f_b}{f_w} \right) c$$

Where,

- $L = 30.5$, median class 31-35
- $N = 38$
- $\sum f_b = 9$
- $f_w = 12$

(c)Median

$$\blacksquare \text{ Median} = L + \left(\frac{\frac{N}{2} - \sum f_b}{f_w} \right) c$$

Where,

- $L = 30.5$, median class 31-35
- $N = 38$
- $\sum f_b = 9$
- $f_w = 12$

(c)Median

$$\blacksquare \text{ Median} = L + \left(\frac{\frac{N}{2} - \sum f_b}{f_w} \right) c$$

Where,

- $L = 30.5$, median class 31-35
- $N = 38$
- $\sum f_b = 9$
- $f_w = 12$
- $c = 5$

(c)Median

$$\blacksquare \text{ Median} = L + \left(\frac{\frac{N}{2} - \sum f_b}{f_w} \right) c$$

Where,

$$\blacksquare L = 30.5, \text{ median class } 31-35$$

$$\blacksquare N = 38$$

$$\blacksquare \sum f_b = 9$$

$$\blacksquare f_w = 12$$

$$\blacksquare c = 5$$

$$\blacksquare \text{ Therefore, Median} = L + \left(\frac{\frac{N}{2} - \sum f_b}{f_w} \right) c$$

(c)Median

$$\blacksquare \text{ Median} = L + \left(\frac{\frac{N}{2} - \sum f_b}{f_w} \right) c$$

Where,

$$\blacksquare L = 30.5, \text{ median class } 31-35$$

$$\blacksquare N = 38$$

$$\blacksquare \sum f_b = 9$$

$$\blacksquare f_w = 12$$

$$\blacksquare c = 5$$

$$\blacksquare \text{ Therefore, Median} = L + \left(\frac{\frac{N}{2} - \sum f_b}{f_w} \right) c$$

$$\blacksquare = 30.5 + \left(\frac{\frac{38}{2} - 9}{12} \right) 5 = 34.67$$

question one

- Find the mode and median of the following distribution below,



question one

- Find the mode and median of the following distribution below,



question one

- Find the mode and median of the following distribution below,

Class interval	Frequency
10-20	3
20-30	2
30-40	10
40-50	4
50-60	5
60-70	2

question Two

- Find the mode and median of the following distribution below,

question one

- Find the mode and median of the following distribution below,

Class interval	Frequency
10-20	3
20-30	2
30-40	10
40-50	4
50-60	5
60-70	2

question Two

- Find the mode and median of the following distribution below,

question one

- Find the mode and median of the following distribution below,

Class interval	Frequency
10-20	3
20-30	2
30-40	10
40-50	4
50-60	5
60-70	2

question Two

- Find the mode and median of the following distribution below,

Class interval	Frequency
4.0-4.2	3
4.3-4.5	8
4.6-4.8	11
4.9-5.1	13
5.2-5.4	12
5.5-5.7	5
5.8-6.0	1

Question three

- From the following distribution



Question three

- From the following distribution



Question three

- From the following distribution



Question three

- From the following distribution

Class interval	Frequency
0.20-0.24	3
0.25-0.29	10
0.30-0.34	12
0.35-0.39	9
0.40-0.44	2
0.55-0.57	5
0.58-0.60	1

find,

- (a) Mean
- (b) Mode
- (c) Median

Question four

- The mean of two samples of sizes 20 and 42 respectively are 24 and 36. Find the combined mean for the two samples

Question five

- The data below are the lengths of 30 plant leaves measured to nearest mm. construct a frequency distribution table having five classes with class interval 10-14, 15-19 etc.

10,20,24,25,30,12,20,21,26,31,13,20,22,27,16,
21,23,28,17,22,22,29,16,21,28,18,17,23,19,15.

Hence,

- (a) Calculate the mean;
 - i) By direct method
 - ii) By assumed mean method
- (b) Mode
- (c) Median



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- Example: Consider the marks: 30%, 70%, 50%, and 90%.
- Range : greatest observation-smallest observation Therefore $90\% - 30\% = 60\%$.



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$$\sum x^2 = N * (\delta)^2 + N * \bar{x}^2$$

$$= 200 * 3^2 + 200 * 48^2 = 1800 + 460800 = 462600$$



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Variance for Grouped data



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- but $\bar{x} = A + \frac{\sum fd}{\sum f}$, where $d = x - A$, $A - \bar{x} = -\frac{\sum f(x - A)}{\sum f} \dots (iii)$

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- If the standard deviation is large, it indicates that there is high degree of dispersion of observations from the mean



Standard deviation for ungrouped data



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■ Standard deviation(S.D)= $\sqrt{\text{Var}(x)} = \sqrt{\frac{\sum(x - \bar{x})^2}{N}}$

Standard deviation for grouped data



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Standard deviation for grouped data

- Standard deviation(S.D) by direct method

$$S.D = \sqrt{Var(x)} = \sqrt{\frac{\sum f(x - \bar{x})^2}{\sum f}}$$



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- Standard deviation(S.D) by Assumed mean method

$$\text{S.D} = \sqrt{\text{Var}(x)} = \sqrt{\frac{\sum fd^2}{\sum f} - \left(\frac{\sum fd}{\sum f}\right)^2}$$



Example

Given the distribution table below;



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Class Interval	Frequency
1-5	5
6-10	10
11-15	12
16-20	6
21-25	8



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Calculate Variance and standard deviation;

- (a) By direct method
- (b) By assumed method



Solution

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Class I.	x	f	c.f	fx	fx^2	$d=x-A$	fd^2	$U = \frac{d}{c}$	fU	fU^2
1-5	3	5	5	15	45	-10	-50	500	-2	-10
6-10	8	10	15	80	640	-5	-40	250	-1	-10
11-15	13	12	27	156	2028	0	0	0	0	0
16-20	18	6	33	108	1988	5	30	150	1	6
21-25	23	8	41	184	4232	10	80	800	2	16

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$$A=13, C=5$$

$$\sum f=41$$

$$\sum fx=543$$

$$\sum fx^2=8889$$

$$\sum fd = 10$$

$$\sum fd^2=1700$$

$$\sum fU = 2$$

$$\sum fU^2 = 68$$

Solution



Solution

(a) Variance and standard deviation by direct method



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$$\text{var}(x) = \frac{\sum fx^2}{\sum f} - \left(\frac{\sum fx}{\sum f} \right)^2 = \frac{1889}{41} - \left(\frac{543}{41} \right)^2 = 41.4$$



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$$\text{Standard Deviation} = \sqrt{\text{Var}(x)} = \sqrt{41.4} = 6.4$$



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The means of two samples of sizes 20 and 42 respectively are 24 and 36. Find combined mean for the two samples.

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The means of two samples of sizes 20 and 42 respectively are 24 and 36. Find combined mean for the two samples.

example

The means of two samples of sizes 18 and 26 respectively are 21 and 23. Find combined mean for the two samples.



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- (iv) d_1 and d_2 are the difference of two mean from combine mean respectively.

The Variance of the combined samples is given by the formula below



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The Variance of the combined samples is given by the formula below

$$\text{Combined Variance } V(x) = \frac{n_1 (\delta_1^2 + d_1^2) + n_2 (\delta_2^2 + d_2^2)}{n_1 + n_2}$$



example

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The mean of two samples of sizes 50 and 100 respectively are 54.1 and 50.3 and the standard deviation are 8 and 7. Find combined mean and the standard deviation of the two samples.

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$$\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2} = \frac{(50 * 54.1) + (100 * 50.3)}{50 + 100} = 51.57$$

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example



example

The mean and standard deviation of a group of 100 observations were found to be 20 and 3 respectively, later on it was found that three observations were incorrect, which are recorded as 21, 21 and 18. Find the mean and standard deviation if the incorrect observations were omitted.

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The mean and standard deviation of a group of 100 observations were found to be 20 and 3 respectively, later on it was found that three observations were incorrect, which are recorded as 21, 21 and 18. Find the mean and standard deviation if the incorrect observations were omitted.

example

Salaries paid to supervisors had a mean of 2500/= with the standard deviation of 2000/=. If all salaries are increased by 2500/=, find the new mean and standard deviation.



example



example

The mean and standard deviation of a set of 100 items were found to be 60 and 10, respectively. At the time of calculation, two items were wrongly taken as 5 and 45 instead of 30 and 20, respectively. Find the corrected standard deviation

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example

If the mean of 5 observations is 4.4 and their variance is 8.24, find the values of the two observations given the three observations are 1, 2 and 6.



example

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Height (cm)	Frequency
131-140	3
141-150	4
151-160	k
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171-180	$k+2$
181-190	5
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- (a) Find the value of k
- (b) By using assumed method determine
 - (i) The actual mean height
 - (ii) The standard deviation of the distribution

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- Therefore the standard deviation must be converted into a relative measure of dispersion for the purpose of comparison.
- The relative measure is known as the coefficient of variation.
- The coefficient of variation is obtained by dividing the standard deviation by the mean and multiply it by 100%. symbolically,



C. V cont....



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- The groups of data for which the **C.V. is greater** indicate that the group is more variable, less stable, less uniform, less consistent or less homogeneous.
- If the **C.V. is less**, it indicates that the group is less variable, more stable, more uniform, more consistent or more homogeneous.



c.v cont....

Example

In two factories A and B located in the same industrial area, the average weekly wages (in rupees) and the standard deviations are as follows

Factory	Average	Standard Deviation	No. of workers
A	34.5	5	476
B	28.5	4.5	524

- Which factory A or B pays out a larger amount as weekly wages?
- Which factory A or B has greater variability in individual wages?



Solution



Solution

- Given Factory A, $N_1 = 476$, $\bar{x}_1 = 34.5$, $\sigma_1 = 5$
Factory B, $N_2 = 524$, $\bar{x}_2 = 28.5$, $\sigma_2 = 4.5$



Solution

- Given Factory A, $N_1 = 476$, $\bar{x}_1 = 34.5$, $\sigma_1 = 5$
Factory B, $N_2 = 524$, $\bar{x}_2 = 28.5$, $\sigma_2 = 4.5$
- i. Total wages paid by factory A; $= 34.5 \times 476 = \text{Rs.}16,422$.
Total wages paid by factory B $= 28.5 \times 524 = \text{Rs.}14,934$.
Therefore factory A pays out larger amount as weekly wages.



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 Therefore factory A pays out larger amount as weekly wages.
- ii. C.V. of distribution of weekly wages of factory A and B are

$$\text{Coefficient of variation (C.V.)}_A = \frac{\sigma_1}{\bar{x}_1} \times 100\% = \frac{5}{34.5} \times 100\% = 14.49\%$$

$$\text{Coefficient of variation (C.V.)}_B = \frac{\sigma_2}{\bar{x}_2} \times 100\% = \frac{4.5}{28.5} \times 100\% = 15.79\%$$

- Therefore Factory B has greater variability in individual wages, since C.V. of factory B is greater than C.V of factory A.

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 Therefore factory A pays out larger amount as weekly wages.
- ii. C.V. of distribution of weekly wages of factory A and B are

$$\text{Coefficient of variation (C.V.)}_A = \frac{\sigma_1}{\bar{x}_1} \times 100\% = \frac{5}{34.5} \times 100\% = 14.49\%$$

$$\text{Coefficient of variation (C.V.)}_B = \frac{\sigma_2}{\bar{x}_2} \times 100\% = \frac{4.5}{28.5} \times 100\% = 15.79\%$$

- Therefore Factory B has greater variability in individual wages, since C.V. of factory B is greater than C.V of factory A.

